



- New chat
- Search chats
- Library
- Atlas
- Projects
- GPTs
- Explore
- Image Reader
- Your chats
- Improve 3DM import code
- Command structure selection
- OpenSCAD parser setup
- Code update TopoShape che...
- Fix TypeError in Python
- Zeeman effect and CO2
- Update function check
- Git ignore .DS_Store
- Improved checkObjShape fu...
- Hull function in OpenSCAD
- UK pensioners full pension d...
- Gitignore .DS_Store files
- FreeCAD API event logging

```
ksloan@Mac importExport3DM %
```

Thanks for sharing the code. I can propose a **cleaner, more robust, and maintainable version** of your `export3DM.py` while keeping the FreeCAD → Rhino3DM export workflow. Here's a detailed improvement plan and the code:

Key Improvements

1. Remove unnecessary switch-case hack
- Python can directly use `if-elif` for type dispatch. The `switch` object is redundant.
2. Class structure cleanup
- Separate responsibilities:
 - `RhinoModel` : handles rhino3dm model creation.
 - `Exporter3DM` : handles FreeCAD object inspection and converts them into Rhino objects.
3. Consistent method names and logging via `FreeCAD.Console.PrintMessage` .
4. Better handling of shapes and surfaces
- Use `Part.BSplineCurve` and `Part.BSplineSurface` directly.
 - Convert `Part::Feature` objects automatically to Rhino curves or surfaces.
5. Error handling
- Avoid code breaking on unsupported types.
 - Print informative messages for unsupported shapes.
6. Remove dead or test code
- Removed unused `processSurfaceUV2/3` , commented-out code, and placeholders.

Improved `export3DM.py`



python

Copy code